

MODUL 8

TRANSACTION PROCESSING, CONCURRENCY, RECOVERY TECHNIQUES

DESKRIPSI TEMA

1. Transaction Processing
2. Concurrency
3. Recovery Techniques

CAPAIAN PEMBELAJARAN MINGGUAN (SUB-CAPAIAN PEMBELAJARAN)

1. Mahasiswa mampu memahami penggunaan operator SAVEPOINT dan ROLLBACK.

PENUNJANG PRAKTIKUM

1. XAMPP
2. Text Editor (Notepad, Notepad++, Visual Studio Code, Sublime, etc.)

LANGKAH-LANGKAH PRAKTIKUM

Petunjuk Praktikum

1. Simpan seluruh hasil jawaban Anda dalam bentuk txt dengan menggunakan text editor kesayangan anda dengan format
Week[Week]_[Kelas]_[NIM]_[NAMA].txt
 Contoh: **Weeko8_A_00000012345_RonnySusanto.txt**
2. Buat database dengan nama ikea dan masuk ke database tersebut
3. Lakukan import database dengan command
SOURCE [path file.sql]
 Contoh: **mysql> SOURCE D:\ikea.sql;**



Tugas Praktikum

1. Tampilkan nama employee dan nama departemen employee tersebut, data diurutkan berdasarkan nama department dan nama employee secara ascending

Name	Department
Mildred Sanders	Bathroom
Ignacio Richardson	Bedroom
Jim Love	Bedroom
Leah Webber	Children's IKEA
Paula Tuckers	Children's IKEA
Dominicus Owen	Cooking
Irma Luna	Cooking
Clay Young	Decoration
Maurice Ford	Decoration
Alfredo Cooper	Dining
Estelle Burton	Dining
Melanie Pierce	Eating
Geranld Phelps	Foods & Beverages
Kevin Padawangi	Foods & Beverages
Martha Mills	Foods & Beverages
Raul Diaz	Foods & Beverages
Christine Ingram	Hallway
Jan Long	Hallway
Audrey Robertson	Laundry
Leticia Hampton	Laundry
Cary Santiago	Lighting
Kendra Jonston	Lighting
Camille Edwards	Living Room
Santago Goodwin	Living Room
Aaron Moreno	METOD Kitchen
Cecil Wade	METOD Kitchen
Jeremy Evan	METOD Kitchen
Philip Burns	METOD Kitchen
Ted Stokes	METOD Kitchen
Wilma George	METOD Kitchen
Leah Caldwell	Outdoor
Sue Price	Outdoor
Ivan Jeremy	Secondary Storage
Jenna May	Secondary Storage
Vellma Carr	Secondary Storage
Alice Fields	Small Storage
Pete Cain	Small Storage
Olga Barrett	Textiles & Rugs
Ruth Mitchell	Textiles & Rugs
Mindy Cannon	Workspaces
Spencer Watts	Workspaces

41 rows in set (0.47 sec)

2. Tampilkan nama employee dan nama manager dari employee tersebut, data diurutkan berdasarkan nama manager dan nama employee secara ascending

Employee Name	Manager Name
Ignacio Richardson	Dominicus Owen
Irma Luna	Dominicus Owen
Jeremy Evan	Dominicus Owen
Jim Love	Dominicus Owen
Leah Webber	Dominicus Owen
Maurice Ford	Dominicus Owen
Olga Barrett	Dominicus Owen
Spencer Watts	Dominicus Owen
Ted Stokes	Dominicus Owen
Alfredo Cooper	Ivan Jeremy
Christine Ingram	Ivan Jeremy
Jenna May	Ivan Jeremy
Leticia Hampton	Ivan Jeremy
Melanie Pierce	Ivan Jeremy
Mindy Cannon	Ivan Jeremy
Pete Cain	Ivan Jeremy
Philip Burns	Ivan Jeremy
Vellma Carr	Ivan Jeremy
Aaron Moreno	Kevin Alexander
Camille Edwards	Kevin Alexander
Cary Santiago	Kevin Alexander
Clay Young	Kevin Alexander
Jan Long	Kevin Alexander
Kendra Jonston	Kevin Alexander
Leah Caldwell	Kevin Alexander
Paula Tuckers	Kevin Alexander
Ruth Mitchell	Kevin Alexander
Wilma George	Kevin Alexander
Alice Fields	Kevin Padawangi
Audrey Robertson	Kevin Padawangi
Cecil Wade	Kevin Padawangi
Estelle Burton	Kevin Padawangi
Martha Mills	Kevin Padawangi
Mildred Sanders	Kevin Padawangi
Raul Diaz	Kevin Padawangi
Santago Goodwin	Kevin Padawangi
Sue Price	Kevin Padawangi

37 rows in set (0.00 sec)

3. Tampilkan nama employee dan setiap order yang ditangani oleh employee tersebut (data orders dipisahkan dengan separator '-'), data diurutkan berdasarkan nama employee

Name	Orders
Alfredo Cooper	OR017 - OR001
Audrey Robertson	OR009 - OR015 - OR020
Camille Edwards	OR002
Cecil Wade	OR011
Clay Young	OR014 - OR018
Jan Long	OR007
Kendra Jonston	OR013
Maurice Ford	OR006
Melanie Pierce	OR005
Mildred Sanders	OR019
Mindy Cannon	OR016 - OR003
Paula Tuckers	OR004
Santago Goodwin	OR008
Vellma Carr	OR010
Wilma George	OR012

15 rows in set (0.02 sec)

4. Tampilkan semua produk beserta jumlah produk yang terjual produk yang ditampilkan hanya produk yang PERNAH TERJUAL MINIMAL 1 kali urutkan tampilan berdasarkan jumlah produk terjual secara descending

Product name	Total sold
BYGEL Container	10
HORDA Open cabinet, wood effect	8
ATTEST Knob	6
EKERÖN Seat/back cushion	4
ASKHOLMEN Balcony table	4
MAXIMERA Divider for high drawer	4
UTRSTA Door damper for hinge	4
ERIKSDAL Handle	3
ANSLUTA Intermediate connection cord	2
ALBERT Shelving unit	2
ANSLUTA Power supply cord	2
HENRIKSDAL Chair cover	2
ELVERDAM Kitchen mixer tap	1
KARLSÖ Gazebo with curtains	1
BLANKETT Handle	1
KLASEN Charcoal barbecue	1
ALGOT Wall upright/shelf/hook	1
DYNING Canopy	1
ÄPPLARÖ Sun lounger	1
FIXA Rear edge trim for worktop	1
TUNHOLMEN Table	1
ATLANT Pop-up drainer	1
BILLSTA Table top	1
BILLSTA Bar table	1
LEIFARNE Seat shell	1
BJURSTA Extendable table	1
ARKELSTORP Sideboard	1
HODDE Rug	1
STORNÄS / KAUSTBY Table and 4 chairs	1

29 rows in set (0.36 sec)

5. Modifikasi query dinomor sebelumnya sehingga produk yang tidak pernah terjual akan tetap ditampilkan dengan total sold 0

* Screenshot dibawah belum mencakup semua baris output

Product name	Total sold
BYGEL Container	10
HORDA Open cabinet, wood effect	8
ATTEST Knob	6
UTRSTA Door damper for hinge	4
EKERÖN Seat/back cushion	4
MAXIMERA Divider for high drawer	4
ASKHOLMEN Balcony table	4
ERIKSDAL Handle	3
ALBERT Shelving unit	2
ANSLUTA Power supply cord	2
ANSLUTA Intermediate connection cord	2
HENRIKSDAL Chair cover	2
TUNHOLMEN Table	1
DYNING Canopy	1
ÄPPLARÖ Sun lounger	1
FIXA Rear edge trim for worktop	1
ATLANT Pop-up drainer	1
STORNÄS / KAUSTBY Table and 4 chairs	1
BJURSTA Extendable table	1
ELVERDAM Kitchen mixer tap	1
LEIFARNE Seat shell	1
ÄLGOT Wall upright/shelf/hook	1
HODDE Rug	1
KÄRLSÖ Gazebo with curtains	1
BLANKETT Handle	1
ÄRKESTORP Sideboard	1
BILLSTA Bar table	1
BILLSTA Table top	1
KLÄSEN Charcoal barbecue	1
ÄPPLARÖ Table/stool section	0
FREDÖN Hammock	0
ÄPPLARÖ Gazebo	0
FIXA Connectors for insert sink/hob	0
FYNDIG Wall cabinet with doors	0
Chicken Hot dog	0
Smoked Salmon	0
FASTBO Wall panel	0
BEKVAM Step stool	0
FYNDIG Base cabinet with doors	0
INGATORP / STEFAN Table and 2 chairs	0
HÄLLÖ Chair cushion	0
METOD Corner wall cab w carousel/glass dr	0
BACKARYD Underframe	0

6. Tampilkan order id, total biaya dari setiap order, beserta nama employee yang menangani order tersebut (price pada orderdetails sudah merupakan hasil kali qty dan price pada product) diurutkan berdasarkan nama employee secara ascending

Order ID	Total	In Charge
OR001	873000	Alfredo Cooper
OR017	475000	Alfredo Cooper
OR020	11779000	Audrey Robertson
OR015	2624000	Audrey Robertson
OR009	1799000	Audrey Robertson
OR002	1380000	Camille Edwards
OR011	2175000	Cecil Wade
OR014	475000	Clay Young
OR018	449000	Clay Young
OR007	5997000	Jan Long
OR013	1700000	Kendra Jonston
OR006	495000	Maurice Ford
OR005	2945000	Melanie Pierce
OR019	8797000	Mildred Sanders
OR016	2149000	Mindy Cannon
OR003	4999000	Mindy Cannon
OR004	498000	Paula Tuckers
OR008	3975000	Santago Goodwin
OR010	799000	Uellma Carr
OR012	449000	Wilma George

20 rows in set (0.02 sec)

7. Perusahaan memutuskan untuk memberikan bonus kepada setiap employee yang telah membawakan keuntungan lebih dari 500.000 (asumsikan perusahaan mengambil keuntungan 12% dari setiap transaksi). Bonus untuk karyawan adalah 5% dari gaji employee tersebut. Tampilkan nama employee, jumlah kontribusi (keuntungan yang sudah diberikan kepada perusahaan), dan juga bonus employee tersebut.
- * angka dibulatkan
- * data diurutkan berdasarkan jumlah kontribusi terbesar.

Employee	Contribution	Bonus
Audrey Robertson	1944240	287500
Mildred Sanders	1055640	300000
Mindy Cannon	857760	300000
Jan Long	719640	287500

4 rows in set (0.00 sec)

8. Tampilkan dalam 1 baris nama employee, jobdesc, dan nama department dengan ketentuan sebagai berikut:
- Penggunaan a/an sesuai tata bahasa inggris.
 - Jika jobdesc adalah restaurant, maka "works as a/an" menjadi "works in"
 - Kolom diberi nama employee information
 - Output tertata rapi sesuai dengan screenshot dibawah
 - Data diurutkan berdasarkan jobdesc, nama department, dan nama lengkap employee
 - Good Luck 😊

Employees information			
Pete Cain	works as a	Communication and Interior Design	in Small Storage Department
Mildred Sanders	works as a	Customer Relations	in Bathroom Department
Maurice Ford	works as a	Customer Relations	in Decoration Department
Alfredo Cooper	works as a	Customer Relations	in Dining Department
Kendra Jonston	works as a	Customer Relations	in Lighting Department
Camille Edwards	works as a	Customer Relations	in Living Room Department
Santiago Goodwin	works as a	Customer Relations	in Living Room Department
Cecil Wade	works as a	Customer Relations	in METOD Kitchen Department
Uellma Carr	works as a	Customer Relations	in Secondary Storage Department
Mindy Cannon	works as a	Customer Relations	in Workspaces Department
Leticia Hampton	works as a	Finance and Business Navigation	in Laundry Department
Christine Ingram	works as a	Human Resource	in Hallway Department
Cary Santiago	works as a	Human Resource	in Lighting Department
Aaron Moreno	works as a	Human Resource	in METOD Kitchen Department
Leah Caldwell	works as a	Human Resource	in Outdoor Department
Alice Fields	works as a	Human Resource	in Small Storage Department
Ignacio Richardson	works as an	IT	in Bedroom Department
Irma Luna	works as an	IT	in Cooking Department
Jeremy Evan	works as an	IT	in METOD Kitchen Department
Spencer Watts	works as an	IT	in Workspaces Department
Jim Love	works as a	Logistics	in Bedroom Department
Leah Webber	works as a	Logistics	in Children's IKEA Department
Ted Stokes	works as a	Logistics	in METOD Kitchen Department
Olga Barrett	works as a	Logistics	in Textiles & Rugs Department
Dominicus Owen	works as a	Manager	in Cooking Department
Kevin Padawangi	works as a	Manager	in Foods & Beverages Department
Ivan Jeremy	works as a	Manager	in Secondary Storage Department
Estelle Burton	works as a	Marketing	in Dining Department
Philip Burns	works as a	Marketing	in METOD Kitchen Department
Jenna May	works as a	Marketing	in Secondary Storage Department
Ruth Mitchell	works as a	Marketing	in Textiles & Rugs Department
Geranld Phelps	works in a	Restaurant (IKEA FOOD)	in Foods & Beverages Department
Martha Mills	works in a	Restaurant (IKEA FOOD)	in Foods & Beverages Department
Raul Diaz	works in a	Restaurant (IKEA FOOD)	in Foods & Beverages Department
Paula Tuckers	works as a	Sales	in Children's IKEA Department
Clay Young	works as a	Sales	in Decoration Department
Melanie Pierce	works as a	Sales	in Eating Department
Jan Long	works as a	Sales	in Hallway Department
Audrey Robertson	works as a	Sales	in Laundry Department
Wilma George	works as a	Sales	in METOD Kitchen Department
Sue Price	works as a	Sales	in Outdoor Department

41 rows in set (0.00 sec)

9. Hilangkan/Disable Autocommit
10. Seorang hacker iseng ingin memainkan database perusahaan tersebut, ia ingin mengubah setiap gaji employee menjadi 0.

Buatlah query untuk mengubah gaji semua pekerjaan menjadi 0.

11. Karena hacker ini ingin melakukan Tindakan lebih maka dia ingin menyimpan progressnya dengan menggunakan savepoint.

Buatlah savepoint dengan nama ikea_gaji_nol.

12. Setelah melakukan savepoint, hacker mengubah setiap nama depan semua employee menjadi Tommy.

Ubahlah nama depan semua employee menjadi "Tommy".

13. Admin database menyadari hal tersebut dan ingin mengembalikan database ke keadaan awal sebelum terserang hacker namun hanya berhasil sampai ke progress yang disimpan oleh hacker.

Tuliskan query untuk melakukan mengembalikan database ke keadaan ikea_gaji_nol!

14. Admin database merasa panik, maka admin melakukan pemulihan sampai ke waktu dimana hacker belum masuk.

Tuliskan query untuk mengembalikan database seperti semula.

REFERENSI